SRI RAM REDDY

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Professional Summary:

- **Data** and **GenAI Engineer** with **5+ years of experience** building production-grade **lakehouse** and **streaming systems** across **AWS, Azure, and GCP,** delivering scalable, real-time data and AI solutions.
- Architected enterprise **data platforms** using **Databricks, Apache Spark, Kafka, and Snowflake,** integrating **GenAI, LLMs,** and **observability frameworks** to optimize performance and compliance.
- Built **Retrieval-Augmented Generation (RAG)** systems using **LangChain, Azure Cognitive Search,** and **OpenAI GPT-4**, enabling contextual query automation and supplier knowledge discovery.
- Delivered **PCI-DSS** and **HIPAA-compliant pipelines** that reduced processing runtimes by 40% and compute costs by 30%, ensuring high availability and robust governance.
- Engineered real-time **fraud detection pipelines** on **AWS EMR, Kinesis,** and **MSK**, enabling sub-second scoring with Terraform-driven autoscaling to meet latency and SLA targets.
- Developed Ford's **LLM-Powered Supplier Quality Assistant** using **Azure OpenAI, Cognitive Search,** and **OpenTelemetry,** improving operational efficiency and system reliability by 70%.
- Designed **ML pipelines** for credit default prediction in **Databricks** using **PySpark, XGBoost,** and **SHAP**, improving model ROC-AUC by 15% and enabling explainable risk insights.
- Migrated 10+ years of **EHR** and **claims data** to **Azure Data Lake Gen2,** enforcing **PHI encryption, Delta Lake ACID compliance**, and real-time analytics dashboards in Power BI.
- Proficient in **Python, PySpark, Scala, SQL, TensorFlow,** and **PyTorch**, delivering optimized, production-grade **machine learning** and **AI pipeline solutions.**
- Automated **data quality** with **Great Expectations**, schema drift detection, and lineage enforcement, improving pipeline reliability and reducing reprocessing cycles.
- Implemented **orchestration** using **Apache Airflow, Terraform,** and **MLflow**, achieving 70% less manual intervention and faster retraining cycles for production ML models.
- Experienced in **prompt engineering, vector databases (FAISS, Pinecone, Pg vector),** and embedding models, building high-accuracy GenAI assistants and enterprise knowledge systems.
- Deployed **containerized APIs** with **FastAPI, Docker,** and **Kubernetes**, sustaining 1K+ RPS throughput with p95 latency <200 ms in live production environments.
- Delivered **analytics solutions** across **fintech** and **healthcare, integrating lineage, dashboards,** and **SLA monitoring** to ensure accuracy and real-time decision insights.
- Collaborated with **DevOps, AI governance**, and **cloud architecture teams** to build secure, compliant, and scalable data ecosystems powering enterprise-level AI workloads.

Technical Skills:

Programming Languages: Python, Scala, Java, R, Shell scripting.

Big Data Technologies: Hadoop (YARN, HDFS, Name Node, Data Node, MapReduce), Apache Spark, Sqoop, Hive, Pig, HBase, Kafka.

GenAI & LLM Technologies: LLMs, GPT, LLaMA, Prompt Engineering, RAG, Embedding Models, Vector DBs (FAISS, Pinecone, Pg vector), LangChain, LangGraph, OpenAI API, HuggingFace, Transformers.

Cloud Platforms: AWS (EC2, S3, Lambda, Redshift), Microsoft Azure (Blob Storage, Data Factory, HDInsight), GCP (BigQuery, Google Cloud Storage (GCS Buckets), Dataproc, Stackdriver, Dataprep, Data Lake, Bigtable).

Data Modeling & Databases: FACT Dimensional Modeling, RDBMS, NoSQL (Mongo DB, Cassandra, Dynamo DB), SQL (Oracle, MySQL, SQL Server, PostgreSQL, Teradata), Distributed Database Systems (Cassandra).

ETL Tools: SQL Server Integration Services (SSIS), Informatica Power Center, Snow SQL.

Database Architectures: OLAP, OLTP, Star Schema, Snowflake Schema.

Data Processing Libraries, BI & Reporting Tools: Pandas, Numpy, Scikit-Learn, Power BI, Tableau, Figma.

Machine Learning & Graph Processing: Spark MLlib, Spark GraphX.

Development Methodologies: Agile, Scrum, Test-Driven Development (TDD).

CI/CD & Version Control: Jenkins, Docker, Concourse, Bitbucket, Git, SVN.

Professional Experience:

***Data and Generative AI Engineer* | Ford Motors Company |Michigan, USA** **Sep 2024– Present**

Project: LLM-Powered Supplier Quality Assistant – Azure GenAI & RAG Platform.

- Architected an enterprise-grade **GenAI assistant** using GPT-4 Turbo with **RAG** on Azure, automating supplier quality queries and **reducing manual lookup time by 70%** across multiple engineering teams.
- Developed a scalable **data ingestion framework** in Azure Databricks to parse and chunk 50K+ supplier PDFs from ADLS Gen2, improving retrieval precision by 60% and reducing latency for search queries.
- Generated vector embeddings using **Azure OpenAI Ada models** and stored them in **Cognitive Search**, enabling **accurate semantic retrieval** and contextual discovery from complex supplier documentation.
- Built multi-turn **prompt orchestration workflows** with LangChain and Azure Functions, enabling adaptive GenAI queries and reducing response inconsistencies by 45% across RAG-driven use cases.
- Containerized and deployed FastAPI microservices on **Azure Kubernetes Service (AKS)** with autoscaling and caching, cutting average response latency by 40% under peak query loads.
- Integrated OpenTelemetry-based **tracing and monitoring pipelines**, improving observability and reliability by 50% and maintaining consistent 99.9% uptime for all production deployments.
- Combined OpenTelemetry metrics with **Azure Monitor** and Key Vault RBAC to **strengthen audit compliance** and achieve ISO 27001-aligned data protection across Toyota's AI workloads.
- Implemented automated **RBAC and access enforcement** through Azure AD and managed identities, **eliminating unauthorized access** incidents and tightening governance for sensitive AI assets.
- Collaborated with DevOps and AI governance teams to optimize **CI/CD pipelines**, achieving **zero-downtime releases** and improved version traceability across all microservice components.
- Delivered a fully compliant, production-ready **LLM platform** that streamlined supplier documentation workflows, reducing operational effort by 55% and boosting team productivity significantly.

***Data Engineer* | Discover Financial Services | Texas, USA** **oct 2023 – Aug 2024**

Project: Credit Default Prediction – End-to-End ML Pipeline on Databricks.

- Built an end-to-end **ETL pipeline** in **Databricks** to process **Discover transactions** from **Snowflake to Delta Lake**, cutting refresh times by 65% and enabling near-real-time analytics.
- Engineered 50+ **behavioral and financial risk features** using **PySpark** and **SQL**, improving overall model **ROC-AUC by 15%** and enhancing predictive performance across backtests.
- Implemented rigorous **data quality checks** with **Great Expectations** and **PySpark validators**, minimizing schema drift and data inconsistencies by nearly 30% in production workflows.
- Optimized **XGBoost** models using **GridSearchCV** with stratified folds and tuned hyperparameters, increasing KS stability by 12% and boosting model generalization accuracy.
- Integrated **SHAP TreeExplainer** with **FAISS** vector retrieval to enable GenAI-driven explainability, surfacing customer-level insights and on-demand risk interpretation.

- Deployed a production-ready FastAPI microservice on **AWS ECS** with **autoscaling** and **ALB routing**, sustaining 1K RPS with p95 latency under 200 ms under heavy load.
- Orchestrated nightly **Airflow DAGs** to automate ETL, retraining, and **MLflow** model registration, reducing manual workflow effort by 70% and ensuring consistent deployments.
- Optimized Spark EMR jobs by enabling **Adaptive Query Execution (AQE)**, broadcast joins, and Z-Ordering, lowering compute runtime by 40% and cost by 25%.
- Developed interactive **Power BI dashboards** on **Azure Synapse** with **DirectQuery**, offering portfolio risk insights, RLS, and explainability roll-ups for credit strategy teams.
- Automated infrastructure provisioning using **Terraform modules** for **Snowflake, S3**, and **ECS**, standardizing environments and reducing setup time from days to minutes.

***Data Engineer* | Medline | Hyderabad, India** **May 2020 – Nov 2022**

Project: Healthcare Data Migration and Analytics Modernization on Azure.

- Migrated over **10 years of healthcare EHR and claims data** to **Azure Data Lake Gen2**, enhancing system **scalability** and **data reliability** by 40% while enabling faster analytics pipelines.
- Developed parallelized **Azure Data Factory (ADF) pipelines** with **watermarking, partitioning**, and **retries,** accelerating ingestion performance by 45% and ensuring end-to-end data freshness.
- Implemented **Delta Lake ACID transactions** and **schema enforcement** to deliver 100% accuracy and consistency across clinical, claims, and operational data reporting.
- Optimized **PySpark ETL transformations** using **broadcast joins**, **partition pruning**, and **Adaptive Query Execution (AQE)**, reducing ETL runtime by 50% and compute costs by 30%.
- Built automated **checksum validation** and **data reconciliation frameworks**, ensuring zero data loss and maintaining consistent data lineage across all environments.
- Enforced **HIPAA compliance** through **PHI masking, AES encryption,** and **dynamic audit policies**, safeguarding sensitive patient data and achieving full regulatory alignment.
- Developed interactive **Power BI dashboards** on **curated Delta tables**, providing real-time clinical insights and reducing report delivery time by 60% for stakeholders.
- Integrated **Azure Key Vault, Private Endpoints**, and **Role-Based Access Control (RBAC)** for secure access management, preventing unauthorized exposure and strengthening data security.
- Collaborated with cross-functional Data Governance and DevOps teams to enhance **CI/CD pipelines**, improving **deployment efficiency** and **system observability**.
- Delivered a scalable, **HIPAA-aligned healthcare data lakehouse** that improved analytical agility, automated reporting, and accelerated **clinical decision-making** across Medline’s operations.

Certifications:

- **AWS Certified Machine Learning Engineer – Associate**
- **AWS Generative AI Applications Professional Certificate**
- **Databricks Certified: Data Engineer Associate**
- **Databricks Certified: Generative AI Engineer Associate**
- **Databricks Certified: Machine Learning Associate**
- **Microsoft Certified: Azure Data Scientist (DP-100)**
- **Microsoft Certified: Azure Data Engineer (DP-203)**

- **Microsoft Certified: Azure AI Fundamentals**

Projects:

Real-Time Fraud Streaming | AWS EMR, Kinesis, MSK, Airflow, Snowflake, Terraform

- Engineered a **real-time fraud detection pipeline** using **PySpark** on **AWS EMR, Kinesis,** and **MSK**, enabling **sub-second scoring** with 99.9% uptime for financial transactions.
- Orchestrated scalable **Apache Airflow DAGs** to manage streaming ingestion, reconciliation, and **Snowflake writes**, ensuring strict **SLA compliance** across all pipelines.
- Implemented **Terraform-based autoscaling** for **EMR clusters**, reducing compute spend by 35% while maintaining **p50/p95 latency targets under 500 ms** consistently.
- Integrated **AWS CloudWatch** and **OpenTelemetry** for real-time metrics, anomaly detection, and error tracing, improving system observability by 40% across workloads.
- Delivered a **production-grade fraud prevention platform** powering **risk analytics dashboards**, improving operational responsiveness by 60% organization-wide.
- Optimized **Kafka topic partitions** and **Spark structured streaming checkpoints**, achieving 35% higher throughput and enhanced fault tolerance under high load conditions.
- Automated data validation and reconciliation workflows with **Great Expectations** and **Airflow sensors,** ensuring **zero message loss** and 100% event accuracy across streams.

Fuzzy Logic Migraine Prediction Using LSTM and Multilayer Perceptron | Python, TensorFlow, Fuzzy Logic

- Developed a hybrid **LSTM and MLP deep learning model** with integrated **fuzzy logic components** in **Python**, improving migraine prediction accuracy and interpretability.
- Achieved 93% accuracy with **LSTM** and **90% with MLP**, proving recurrent architectures outperform dense models on temporal healthcare data for early migraine detection.
- Designed fuzzy membership and rule-based inference logic to manage uncertainty, enhancing decision transparency and increasing clinician confidence in predictions.
- Implemented end-to-end **TensorFlow pipelines** for model training, validation, and tuning, reducing convergence time and improving model reproducibility by 25%.
- Integrated **scikit-fuzzy** and **NumPy** modules to build interpretable fuzzy systems, allowing granular thresholding and contextual mapping of patient-specific signals.
- Applied **time-series preprocessing** using **pandas** and **Signal Smoothing** to handle noisy input features, improving data quality and predictive performance.
- Optimized model deployment using **ONNX** and **TensorFlow Lite**, reducing model inference time by 35% for real-time clinical prediction readiness.
- Delivered an interpretable **AI-driven migraine prediction platform**, supporting healthcare analytics research and early intervention strategies for patient well-being.

Education:

Master of Science in Information Technology & Management — **Webster University**